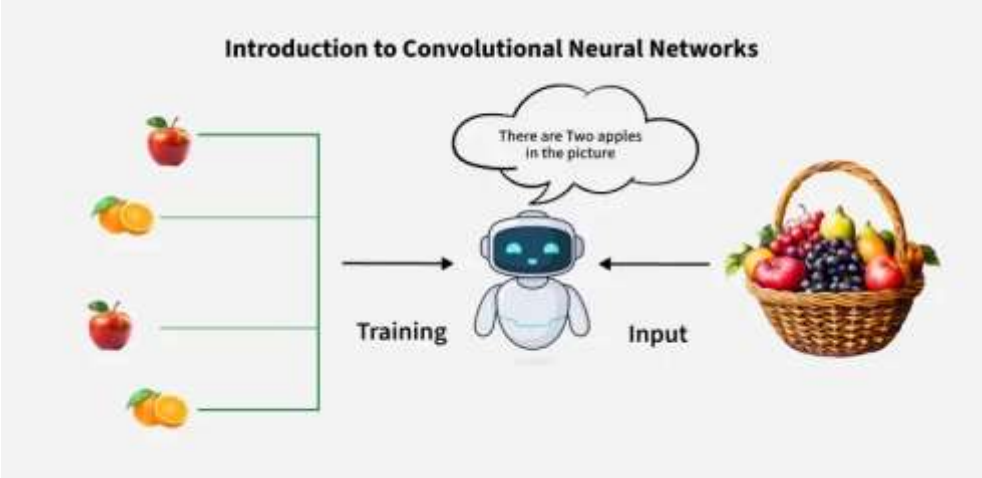
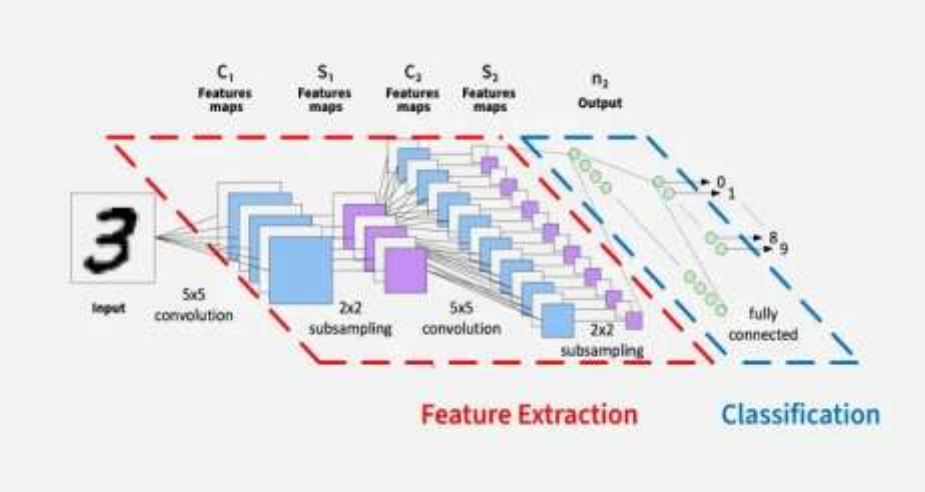
Day – 17 Convolutional Neural Network

Convolutional Neural Networks (CNNs) are deep learning models designed to process data with a grid-like topology such as images. They are the foundation for most modern computer vision applications to detect features within visual data.



Key Components

- Convolutional Layers: These layers apply convolutional operations to input images using filters or kernels to detect features such as edges, textures and more complex patterns. Convolutional operations help preserve the spatial relationships between pixels.
- Pooling Layers: They downsample the spatial dimensions of the input, reducing the computational complexity and the number of parameters in the network. Max pooling is a common pooling operation where we select a maximum value from a group of neighboring pixels.
- Activation Functions: They introduce non-linearity to the model by allowing it to learn more complex relationships in the data.
- Fully Connected Layers: These layers are responsible for making predictions based on the high-level features learned by the previous layers. They connect every neuron in one layer to every neuron in the next layer.



Working of CNN:

1. **Input Image**: CNN receives an input image which is preprocessed to ensure uniformity in size and format.
2. **Convolutional Layers**: Filters are applied to the input image to extract features like edges, textures and shapes.
3. **Pooling Layers**: The feature maps generated by the convolutional layers are downsampled to reduce dimensionality.
4. **Fully Connected Layers**: The downsampled feature maps are passed through fully connected layers to produce the final output, such as a classification label.
5. **Output**: The CNN outputs a prediction, such as the class of the image.

Training a Convolutional Neural Network

CNNs are trained using a supervised learning approach. This means that the CNN is given a set of labeled training images. The CNN learns to map the input images to their correct labels.

The training process for a CNN involves the following steps:

1. **Data Preparation:** The training images are preprocessed to ensure that they are all in the same format and size.
2. **Loss Function:** A loss function is used to measure how well the CNN is performing on the training data. The loss function is typically calculated by taking the difference between the predicted labels and the actual labels of the training images.
3. **Optimizer:** An optimizer is used to update the weights of the CNN in order to minimize the loss function.

4. **Backpropagation:** Backpropagation is a technique used to calculate the gradients of the loss function with respect to the weights of the CNN. The gradients are then used to update the weights of the CNN using the optimizer.

How to Evaluate CNN Models

Efficiency of CNN can be evaluated using a variety of criteria. Among the most popular metrics are:

- **Accuracy:** Accuracy is the percentage of test images that the CNN correctly classifies.
- **Precision:** Precision is the percentage of test images that the CNN predicts as a particular class and that are actually of that class.
- **Recall:** Recall is the percentage of test images that are of a particular class and that the CNN predicts as that class.
- **F1 Score:** The F1 Score is a harmonic mean of precision and recall. It is a good metric for evaluating the performance of a CNN on classes that are imbalanced.